THE SAFETY OF WILD BOAR (SUS SCROFA) MEAT HUNTED IN DIFFERENT EUROPEAN COUNTRIES

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Introduction

The wild boar (Sus scrofa) is a popular game animal worldwide. Boar populations in Europe and Asia are numerous and widespread. However, these animals pose numerous problems: they damage agricultural crops, spread infectious diseases that are dangerous to pigs (e.g., African swine fever, ASF), encroach on human settlements, cause traffic accidents, and are an increasing threat to humans. Attempts to reduce the number of wild boars have a low success rate, since, thanks to their rapid reproduction, their herds quickly recover after periods of intensive shooting. In addition, they perform well even in hostile environments and readily colonize new areas. Hunting and systematic culling of wild boars are aimed at reducing their numbers and population density. In 2019-2023, the annual harvest of wild boar carcasses ranged from 221,000 to 420,000. After mandatory veterinary inspection, wild boar meat can be a more environmentally friendly, low-cost alternative to pork. Considering the findings of various authors, the rules for marketing and consuming this meat should be followed. Only meat from wild boar carcasses that have undergone mandatory testing for *Trichinella spiralis* and *Alaria alata* should be consumed. The consumption of wild boar offal, especially liver and kidneys, should be strictly avoided due to the accumulation of heavy metals and radioactive contamination in these organs. Venison is an organic food, free of antibiotics and growth promoters. However, due to the risks presented, it requires special treatment and veterinary control.

Keywords: wild boars, meat, safety

Results

Table 1 Content of heavy metals and arsenic in the muscle tissue of wild boars hunted in various European countries.

Characteristic	European countries						
	Poland mg/kg	Poland mg/kg	Slovakia mg.kg		Slovakia µg/g w.w.	Hungary mg/kg w.w	Romania mg/g
Hunting season	IX 2011 – XII 2013	2010-2011	2014/2015 – 2017/2018,		XI/XII 2009 and 2010	VIII/IX2018	winter 2010–2021
Hunting area	Upper Silesia, Turoszow	Podkarpacie Subcarpathia	Tatra National Park	Zemplin	region of Nitra and Topolcianky	Central Transdanubia Region	North-Eastern Romania
Arsenic (As)	nd	nd	0.06-2.94	0.001-0.02	nd	<0.5	nd
Cadmium (Cd)	0.001 - 0.140	0.01-0.06	0.003-2.63	0.024-1.53	0.043-0.373	<0.05	0.01-0.98
Lead (Pb)	0.002 - 0.130	0.06-0.18	0.009-1.894	0.001-0.36	0.039-61.3	0.22 - 0.36	0.025-0.302
Mercury (Hg)	0.001 - 0.035	nd	0.02-0.871	0.001-0.137	0.000-0.251	<0.5	nd
Chromium (Cr)	nd	nd	nd	nd	nd	nd	nd

Characteristic	European countries						
	Croatia (mg/kg)	Croatia mg/kg	Austria mg/kg wet mass	Italy mg kg^{-1} ww	Italy mg/kg w.w.	Italy Sardinia mg/kg dw	Turkey mg/kg
Hunting season	2008/2009	nd	2012-2013	XI 2005 - I 2006	nd	2022–2023	IX 2013 - II 2014
Hunting area	between the Sava and Drava	Papuk Nature Park	Eastern Austria	Viterbo Province	Molise - Bagnoli del Trigno and Roccavivara	northern Sardinia	Kırıkkale province
Arsenic (As)	nd	0.012-0.286	0.012± 0.011	nd	nd	nd	nd
Cadmium (Cd)	0.0002–0.428	0.004-0.672	0.002± 0.001	0.031–0.381	0.001	nd	0.51±0.02
Lead (Pb)	0.001–1.01	0.03-9.1	0.015± 0.017	0.080 – 0.226	0.011	0.011-85.76	0.44±0.17
Mercury (Hg)	0.001–0.036	0.005-0.103	0.008± 0.005	nd	nd	nd	nd
Chromium (Cr)	nd	nd	0.007± 0.006	0.069 – 0.692	0.066	nd	1.35±0.59

nd - no data

Table 2 Content of cesium ^{137}Cs and cesium ^{134}Cs in the tissues of wild boars hunted in various European countries (Bq/kg fresh meat).

Characteristics	European countries			
	Poland	Serbia	Slovakia	Czech Republic
Hunting season	2013/2014 - 2016/2017	May 1st - January 31st in 2020 and 2021	2017 - 2019	December 2012 - December 2019
Hunting area	Lower Silesia, Opole, Subcarpathia	four regions of Serbia	various locations in Slovaki	Novohradské (Gratzen)
Cesium^{137}Cs (Bq/kg)	<1.18 - 11.47	0.05–0.38 Bq/ kg North Serbia, 0.21–5.18 Bq/ kg Eastern Serbia	0.40 - 37.2	981 Bq·kg^{-1} Median 358 Maximum 14252
Cesium ^{134}Cs (Bq/kg)	<1.06	0.01–0.07 Bq/ kg - North Serbia, 0.02–1.11 Bq/ kg - Eastern Serbia	nd	nd

Characterristics	European countries			
	Belarus	Germany	Italy	Italy
Hunting season	1991-2008	2019–2021	nd	2014and2016
Hunting area	Belarusia Polesie.	Bavaria	Reggio Calabria	Chisone/Germanasca Valley and PelliceValley districts (Piedmont)
Cesium ^{137}Cs (Bq/kg)	0.5-661 kBq/kg in the Alienation Zone, 0.3- 105 kBq/kg in the permanent control zone and 0.1-2.4 kBq/kg in the periodic control zone	0.37-14 kBq·kg^{-1}	0.06 - 2.14 Bq kg^{1}	2014 liver 0.0-39.7 Bq/kg; 2016 liver 3.1-113.3 Bq/kg Kidney 4.5-192.0 Bq/kg
Cesium ^{134}Cs (Bq/kg)	nd	nd	nd	nd

nd - no data

Table 3 Content of Polychlorinated Biphenyles Congeners (PCB) and pesticides in the fat tissue of wild boars (µg.kg^{-1} fat) (n= 12)

Polychlorinated Biphenyles Congeners	Found	Not found
PCB 101	0	12
PCB 118	0	12
PCB 138	1 (Lublin - 49 µg/kg fat)	11
PCB 153	3 Lublin 99.9±25.0 µg/kg fat Lublin 96.0±18.4 µg/kg fat Lower Silesia 9.5±1.4µg/kg fat	9
PCB 180	1 Lublin 112 µg/kg fat	11
PCB 28	1 Lublin 297.2±59.4 µg/kg fat	11
PCB 52	1 Lublin 300.1±39.0 µg/kg fat	11
DDE	6 (25.0 - 973.0 µg/kg fat)	6
p,p'- DDE	4 (11.5±4.1 - 107.8±38.8)	8
p,p'- DDT	1 (38.2±13.7)	11

Conclusion

Data from numerous European studies revealed a consistent, organ-specific pattern of heavy metal accumulation in wild boars. Muscle tissue generally contained lower concentrations of toxic elements than internal organs; however, lead (Pb) emerged as the most critical contaminant in meat intended for human consumption. In muscle tissue, Pb concentrations ranged from trace levels up to extremely high values, reaching 61.3 mg/kg wet weight in Hungary and 85.76 mg/kg dry weight in Sardinia, far exceeding EU maximum permissible limits. These elevated concentrations are largely attributed to secondary contamination from lead-based ammunition rather than environmental exposure alone. Cadmium (Cd) concentrations in muscle were typically lower but still reached 2.63 mg/kg in Slovakia and 0.98 mg/kg in Romania, with values in several regions approaching or exceeding regulatory thresholds, particularly in older animals.

In addition to heavy metals, wild boars have been shown to accumulate radiocaesium (137Cs), with concentrations ranging from background levels to extreme values exceeding 14,000 Bq/kg, particularly in regions affected by historical radioactive fallout, such as the Novohradské Mountains. Furthermore, residues of chlorinated hydrocarbons, pesticides, per- and polyfluoroalkyl substances (PFAS), and emerging contaminants, including microplastics, were detected, reflecting omnivorous feeding behavior, extensive home ranges, and frequent foraging in agricultural, urban, and waste-affected environments.

These results demonstrate that while wild boar meat can represent a valuable and sustainable food resource, it is also characterized by a uniquely complex contamination profile. Safe consumption requires mandatory testing for *Trichinella spiralis* and *Alaria alata*, strict exclusion of offal—especially kidneys and liver—from the food chain, regular monitoring of heavy metals and radionuclides, and the reduction or elimination of lead-based ammunition. From a public health and One Health perspective, wild boar meat should be classified as a high-risk game product, and its increasing availability should be accompanied by enhanced surveillance, evidence-based regulation, and clear consumer guidance.

Literature:
Migdał, W., Radović, Č., Živković, V.B., & Migdał, Ł. (2026). The safety of wild boar (Sus scrofa) meat hunted in different European countries. Scifood, vol. 20, 74-91. https://doi.org/10.5219/scifood.94